

YASHRAJ SAH

Agentic AI | Generative AI | LLM Engineer | Multi-Agent Systems | RAG | Python

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PROFESSIONAL SUMMARY

B.Tech Computer Science & Engineering (AI & ML) graduate with hands-on experience building Agentic AI, Generative AI and LLM-powered applications. Completed an Agentic AI / GenAI internship with Tata Consultancy Services (TCS) under the NPI Value Stream, designing a multi-agent, RAG-based enterprise decision-support platform. Skilled in multi-agent system design, retrieval-augmented generation (RAG), LLM application development and intelligent automation, with additional project experience spanning supply-chain decision intelligence and natural-language-to-CAD generative AI. Strong Python backend/full-stack developer (FastAPI, LangChain, LangGraph, React/Next.js) and Claude Certified Developer – Foundations. Looking to contribute to Agentic AI, GenAI and AI/ML Engineering roles.

TECHNICAL SKILLS

Agentic AI & GenAI: Multi-Agent Systems, AI Agents, LLM Applications, RAG (Retrieval-Augmented Generation), LangChain, LangGraph, Prompt Engineering, Semantic Search, Vector Search, AI Automation, Decision Intelligence

LLMs & AI: OpenAI, Groq, LLaMA 3 / 3.3, Ollama, Hugging Face Transformers, Embeddings, Sentence Transformers

Programming: Python, SQL, JavaScript / TypeScript

Backend & APIs: FastAPI, Flask, REST APIs, API Integration, JWT Authentication, OAuth, Node.js, Pydantic

Data & ML: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Forecasting

Frontend: React.js, Next.js, Tailwind CSS, Streamlit, HTML/CSS

Databases & Search: PostgreSQL, Supabase, MySQL, FAISS, Google Sheets API

Automation & Tools: Workflow Automation, Webhooks, Event-Driven Systems, Selenium, Pytest, Git, GitHub, Docker, Postman, Google Workspace APIs

Cloud & DevOps: AWS Fundamentals, CI/CD Concepts

INTERNSHIP EXPERIENCE

Agentic AI / GenAI Intern — Tata Consultancy Services (TCS)

Mar 2026 – Aug 2026

NPI (New Product Initiation) Value Stream | Digital Garage Innovation Lab – Rapid Experimentation & Value Creation

- **Project:** Enterprise Change Impact Analysis Agent — an Agentic AI platform that automates impact analysis for Engineering Change Requests, which previously required engineers to manually gather data from PLM, SAP/ERP, QMS, Manufacturing Systems, BOM/EBOM repositories, Engineering Knowledge Bases and Supplier records.
- Designed a **multi-agent architecture** with specialized agents — Engineering, Manufacturing, Supply Chain, Validation/Compliance, Financial Impact and Change Review Board (CRB) Decision — to assess and recommend change-request outcomes.
- Built the end-to-end agent workflow and orchestration logic, integrating **LLM reasoning** (OpenAI / Groq) with retrieval pipelines for grounded, enterprise-context-aware decision recommendations.
- Developed a **RAG pipeline** using LangChain, FAISS and Sentence Transformer embeddings for semantic search over enterprise engineering documentation.
- Diagnosed and resolved retrieval quality issues (irrelevant matches from large chunks, low similarity thresholds and noisy historical documents) by implementing **metadata filtering, optimized chunking, similarity-threshold tuning, hybrid keyword + semantic search, and a retrieval quality validation layer** — improving grounding and reducing hallucination risk.
- Built backend services and APIs with FastAPI and a Streamlit interface for internal demonstration; used Pandas for enterprise data processing.
- Project material estimated a target of **~60-80% reduction in manual impact-analysis effort** with faster engineering decision cycles and improved consistency/traceability (internal project estimate, not an independently audited production metric).

AI/ML Intern — Western Heat & Forge Pvt. Ltd.

Jul 2025 – Jan 2026

AI/ML Engineering | Workflow Automation | Full-Stack Development

- Built production-ready automation systems across HR, Production, Marketing, Security and RFQ departments, serving more than 200 users across multiple departments.
- Developed scalable FastAPI and Flask backend services with webhook validation, retry mechanisms and structured logging.
- Designed reusable automation modules integrated with WhatsApp Business API, email services and Google Workspace.
- Built a WhatsApp Taskbot API enabling intelligent task creation, assignment and tracking, with two-way Google Sheets synchronization.
- Developed an AI-powered HR induction chatbot to streamline employee onboarding, with JWT-based secure authentication.
- Automated marketing/RFQ cost estimation using historical pricing data, cutting processing time from days to minutes.
- Improved overall workflow efficiency by automating manual business processes and reducing turnaround time significantly.

SELECTED PROJECTS

SmartChain Nexus — Agentic AI Supply Chain Decision Intelligence System

Python, FastAPI, LangGraph, Groq LLaMA 3.3, Supabase PostgreSQL, Next.js, React, Tailwind CSS, Pandas | github.com/Thenameisyashraj1707/SmartChain-Nexus-Agentic-AI-Supply-Chain-Decision-Intelligence-System

- Built a multi-agent supply-chain decision-intelligence / control-tower system that answers natural-language supply-chain queries and generates operational recommendations.
- Implemented a **DecisionLLMAgent** and specialized analysis pipelines covering supply-chain risk assessment, inventory analysis, supplier analysis, demand intelligence, reorder optimization, transfer optimization, slotting optimization and forecasting.
- Evolved the system from static CSV-based prototyping to a Supabase-backed intelligence layer, exposed via a FastAPI backend with a Next.js/React dashboard.

AutoDev — Autonomous AI Software Engineer

Agentic AI, LLMs, Python, Multi-Agent Architecture

- Designed a multi-agent software engineering system with Planner, Architect, Coding, Execution, Recovery, QA and Security agents to automate code generation, execution and validation.
- Implemented self-healing repair loops that diagnose failures, debug and re-validate code automatically, reducing manual intervention in the development loop.

Generative AI Based Automated 3D CAD Model Generator

Final-Year Project + Industry Project | Python, Ollama, LLaMA 3, CadQuery, OpenCascade, Pandas, CSV, Tkinter |

github.com/Thenameisyashraj1707/Generative-AI-Based-Automated-3D-CAD-Model-Generator

- Built a Generative AI-powered natural-language-to-CAD system that converts engineering design requirements expressed in plain English into parametric CAD models.
- Used a local LLM (LLaMA 3 via Ollama) to extract part type, dimensions and pressure requirements from natural-language input, mapping them to engineering parameters.
- Automated parametric geometry generation for API 6A components (bonnets, flanges, spools) using CadQuery/OpenCascade, with STEP export for downstream CAD/NX workflows.
- Designed for local/offline-capable operation; delivered as both the final-year academic project and an industry-facing engineering project.

WhatsApp Taskbot API — Workflow Automation

Python, Flask, Google Workspace, Webhooks, Meta WhatsApp Business API

- Built an intelligent WhatsApp chatbot for task creation, assignment, tracking and notifications, with secure two-way Google Sheets synchronization.

Visitor Management System

React.js, TypeScript, Tailwind CSS

- Developed a digital visitor management platform with QR-based check-in/check-out, webcam capture and Google Drive storage, managing 1,000+ monthly visitor entries.

EDUCATION

Vishwakarma Institute of Information Technology (VIIT), Pune

2022 – Apr 2026

B.Tech. – Computer Science & Engineering (Artificial Intelligence & Machine Learning), CGPA: 7.87/10

SNBP Jr. College, PCMC

2020 – 2022

Higher Secondary Certificate (Science), Percentage: 85.33%

CERTIFICATIONS

- **Claude Certified Developer – Foundations** — Anthropic / Credly (Issued: Aug 31, 2026; Expires: Aug 31, 2027)
- Complete Generative AI with LangChain & Hugging Face
- Complete Python Bootcamp: Zero to Hero
- IBM AI Engineering
- IBM DevOps for AI
- Cisco CCNA
- Infosys Internship Certificate
- Tata Consultancy Services (TCS) Internship Certificate

ACHIEVEMENTS

- Secured 2nd Rank in Sci-Tech Exhibition.